

MATTHEW CURRAN

ELECTRICAL ENGINEERING STUDENT - MEMORIAL UNIVERSITY OF NEWFOUNDLAND

1-709-697-2605 | mvcurran@mun.ca | St. John's, NL

RELEVANT EXPERIENCE

Valiant Aerotech Design Team, Electrical Engineering Team

St. John's | Jan 2026 - Present

- Designed and developed PCB schematics and circuit layouts using KiCad for aerospace focused embedded systems, supporting reliable integration of sensors, power distribution components, and onboard communication hardware.
- Performed precision soldering, hardware assembly, and electrical testing of prototype systems, ensuring robust connections, component reliability, and successful integration of avionics and mission critical subsystems.
- Configured and optimized flight and mission parameters within Mission Planner while contributing to embedded software development in C++ and C to support autonomous control, telemetry, and system performance for aerospace applications.

GovNL Provincial Radio Communications Office, Electrical Engineering Student

St. John's | Jan - April 2026

- Designed and analyzed the power distribution, wiring architecture, and equipment configuration for a provincial emergency interoperability dispatch centre, ensuring reliable integration of multi-agency communications systems.
- Implemented Radio over IP (RoIP) gateway infrastructure, performing hardware assembly (soldered connections and interface wiring) and software configuration to bridge radio networks with IP-based communications.
- Evaluated system power consumption and equipment logistics, supporting infrastructure planning and documentation for critical provincial emergency communications systems.

Newfoundland and Labrador Hydro, Transmission and Distribution Engineering Student

St. John's | May - Aug 2025

- Digitized existing transmission lines into three dimensional models to critically analyze transmission engineering problems, evaluating alternatives and recommending course of action (using PLS-CADD, JDEdwards, and ArcGIS software).
- Proposed a new 400 kilometer transmission line spanning across south-eastern Labrador that would decrease provincial diesel power generation and long term costs, complete with a PLS-CADD digital model and nine-figure cost estimation.
- Maintained accuracy of engineering drawings by updating outdated part numbers and supporting BOM management to align design documentation with production.

GovNL Department of Tourism, Provincial Historic Sites Tour Guide

St. John's | May - Aug 2024 | June - Aug 2023

- Independently delivered dynamic and historically accurate tours in both English and French, engaging diverse audiences across multiple historic sites.
- Managed site operations including cash handling, retail transactions, and end-of-day reporting, ensuring smooth daily openings and closings.
- Contributed to the site's educational mission by refining tour content through self-led research and visitor feedback.

EXTRACURRICULARS and VOLUNTEERING

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| Mathematics Tutor | 2024 - Present |
| Assistant Coach of Holy Heart of Mary's High School Men's Volleyball Team | 2024 - Present |
| Men's Intramural Volleyball League | 2024 - Present |
| Delta Kappa Epsilon Fraternity - University of British Columbia | 2023 - 2024 |
| Stick-It to Cancer Vancouver Charity Event Coordinator | 2023 - 2024 |
| Captain of Holy Heart of Mary's Men's Volleyball Team | 2022 - 2023 |

EDUCATION

Memorial University of Newfoundland and Labrador

Sep 2024 - May 2029

- Bachelors of Engineering, Electrical

University of British Columbia

Sep 2023 - Apr 2024

- Bachelors of Applied Science and Engineering (completed first year studies)

SKILLS

Excellent written and oral communication, critical thinking skills, safety acumen, leadership, class 5 driver's license, emergency first Aid/CPR A/AED, KiCad, Microsoft Suite, CAD, Python, C, C++.